

Supplementary Informaion

Supplementary Table 1 | Overview of additional MRI sequences and patient information utilized in radiological diagnosis in the primary test dataset and external test dataset. Extra MRI sequences refer to imaging techniques beyond standard T1, T1c, T2, and FLAIR. These extra sequences include Diffusion-Weighted Imaging (DWI), Diffusion Tensor Imaging (DTI), Perfusion-Weighted Imaging (PWI), Magnetic Resonance Angiography (MRA), Magnetic Resonance Spectroscopy (MRS), Susceptibility-Weighted Imaging (SWI), Magnetic Resonance Venography (MRV), Blood Oxygen Level Dependent Imaging (BOLD), Computed Tomography (CT), and Apparent Diffusion Coefficient (ADC). Additional patient information includes access to medical history and prior radiology imaging and associated findings, offering further clinical context for diagnostic decision-making.

Dataset	Total Number	Extra Modality Percentage	Extra Prior Medical History of Pathological diagnosis	Overall Extra Information Percentage
Primary test dataset	3,797	39.5%	19.3%	58.8%
External dataset	1,309	56.1%	12.7%	68.8%

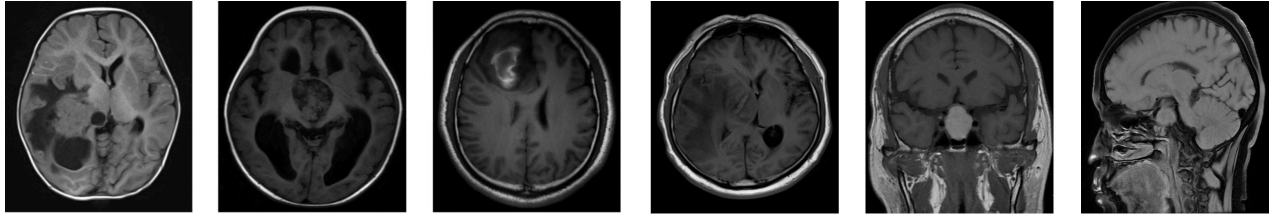
Supplementary Table 2 | Summary of MRI scanner manufacturers, models, and scanning parameters used across all medical centers.

Scanning Protocol	MRI Scanner Manufacturer			
	GE	Siemens	Philips	UIH
Model	Discovery MR750w, Signa HDxt, Signa Excite	Skyra, Prisma, Avanto, Syngo	Ingenia, Ingenuit	uMR 660
Field strength	1.5 T/ 3.0 T	1.5 T/ 3.0 T	3.0 T	1.5 T
Flip angle	1°, 8°, 12°, 15°, 18°, 19°, 20°, 25°, 30°, 40°, 60°, 70°, 90°, 111°, 112°, 125°, 142°, 155°, 160°	7°, 8°, 9°, 10°, 15°, 20°, 24°, 25°, 27°, 30°, 40°, 50°, 60°, 70°, 80°, 90°, 120°, 122°, 140°, 145°, 150°, 160°, 180°	8°, 10°, 15°, 17°, 18°, 20°, 30°, 40°, 45°, 70°, 75°, 80°, 90°, 100°	10°, 12°, 15°, 20°, 25°, 54°, 60°, 70°, 72°, 90°, 120°, 130°, 140°, 150°
Pixel Spacing Range (mm^2)	0.17 × 0.17 - 3.5 × 3.5	0.23 × 0.23 - 4.7 × 4.7	0.21 × 0.21 - 1.8 × 1.8	0.26 × 0.26 - 1.4 × 1.4
Slice Thickness Range (mm)	0.6 - 278	0.47 - 60	0.75 - 398	0.5-10

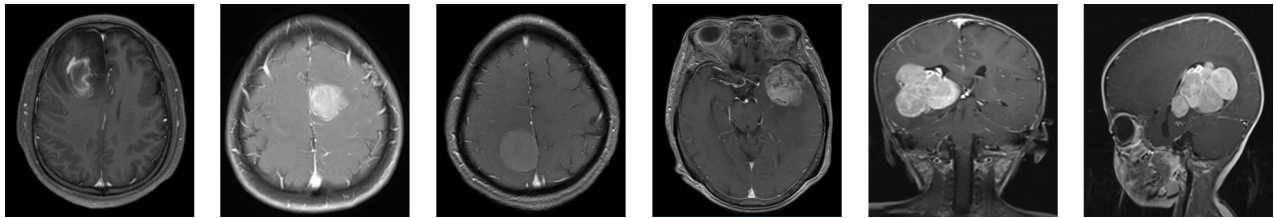
Supplementary Table 3 | Abbreviation of 12 major brain tumors in WHO CNS5.

Acronym	Original Tumor Type Name
GGN	Glomas, glioneuronal tumors, and neuronal tumors
MET	Metastasis
GCT	Germ cell tumor
MEN	Meningioma
TSR	Tumor of the sellar region
MNM	Mesenchymal, non-meningothelial tumor
CPN	Cranial and paraspinal nerve tumor
CPT	Choroid plexus tumor
HEM	Hematolymphoid tumor
EMB	Embryonal tumor
PIN	Pineal tumor
MEL	Melanocytic tumor

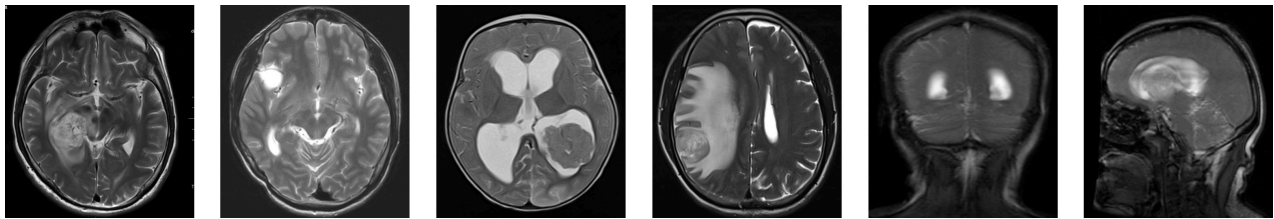
a. T1 Comparison



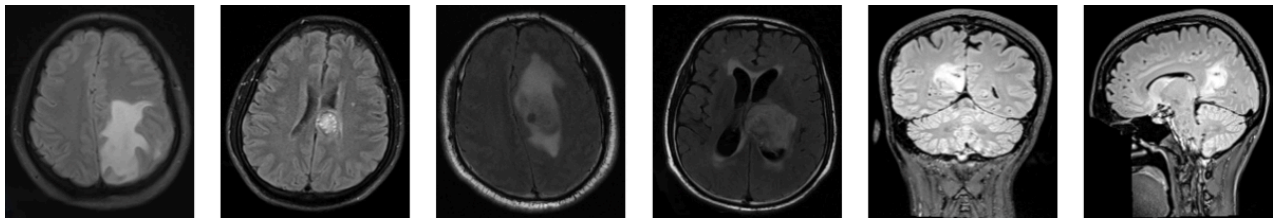
b. T1c Comparison



c. T2 Comparison

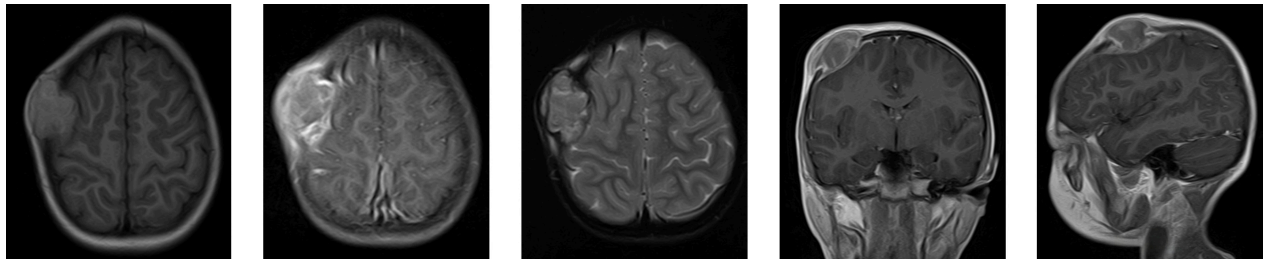


d. FLAIR Comparison

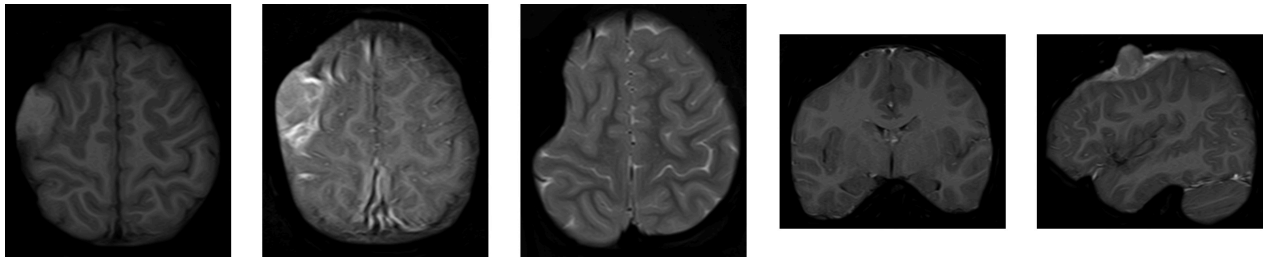


Supplementary Fig. 1 | Visualization of MRI images acquired using different imaging parameters across MRI sequences. Shown are T1-weighted (T1), contrast-enhanced T1-weighted (T1c), T2-weighted (T2), and T2 FLAIR (FLAIR) sequences. For each sequence, four axial images obtained using different imaging parameters and MRI devices are presented. Additionally, one representative sagittal and one coronal view acquired under a specific parameter setting are included for each sequence.

a. Original MRI image



b. MRI image after skull-stripping



Supplementary Fig. 2 | Comparison of unprocessed and skull-stripped MRI images. Representative examples of MRI scans before and after skull-stripping. While conventional skull-stripping may aid certain tumor analyses, it is not universally appropriate. Crucially, lesions located in regions like the pia mater or invading the cranial vault can be inadvertently removed during skull stripping, leading to the loss of critical anatomical and pathological information – a risk our approach deliberately avoids.